Lab 03 – Principal Components Analysis (PCA)

Analytics and Knowledge Discovery, FH Salzburg. Summary of the three exercises PC1 (superheroes), PC2 (MNIST) and PC3 (MovieLens).

The analysis recipe

• **Data understanding / EDA:** rows, columns, kinds of entries, missing values and correlations (PCA only finds *linear* redundancy).

• **Normalize?** Standardize (z-score) if the variables have different units or scales. Don't standardize if all variables share the same meaning and range.

• **PCA:** rotate to the directions of largest variance. Every PC is a linear combination of the variables, and its coefficients are the *loadings*.

• **Choose the number of dimensions:** from the cumulative explained variance (rule of thumb 90 – 95 %); the eigenvalue > 1 rule (Kaiser) is an alternative for standardized data.

• **Interpret:** look at the loadings of the first PCs and at the biplot (points = observations, arrows = variables).

Overview

	PC1 Superheroes	PC2 MNIST	PC3 Movies
Data	58 heroes × 7 attributes	10 000 images × 784 pixels	610 users × 9 724 movies
Normalize?	Yes – units and scales differ	No – same range 0–255, only /255	No column scaling; center each user, missing → 0
PC1+PC2	57 %	17 %	8 %
90 % needs	5 PCs	86 PCs	≈ 250 PCs (max. 609)
2-D view	useful, interpretable	only 0 and 1 stand apart	of limited use

PC1 – Superhero data

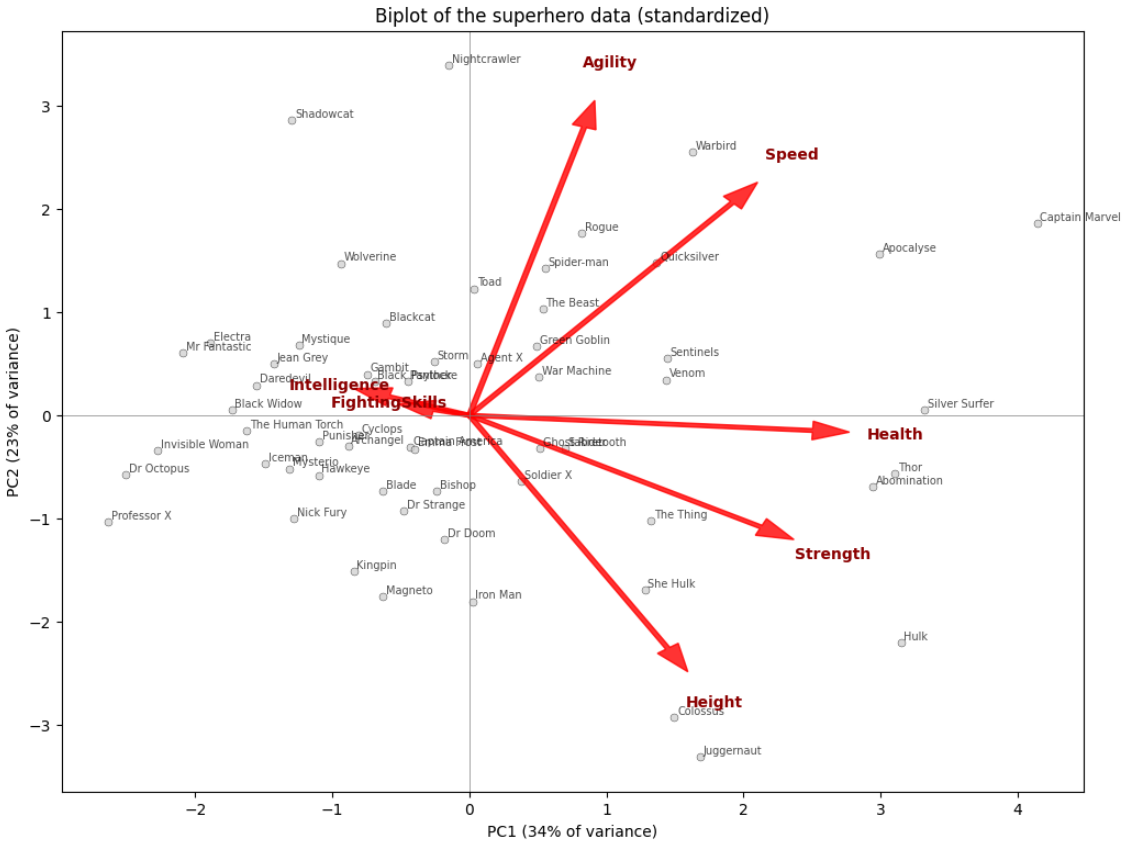
• 58 Marvel characters with Height (cm), Intelligence, Strength, Speed, Agility, FightingSkills and Health. 11 missing values were filled with the column median.

• **Standardization is needed:** without it PC1 explains 95 %, but PC1 is just *Health* (loading 0.99), which has by far the largest variance.

• PC1 ≈ 0.59·Health + 0.50·Strength + 0.45·Speed + 0.34·Height + 0.19·Agility − 0.11·FightingSkills − 0.18·Intelligence.

• Explained variance is 34 / 23 / 15 / 14 / 7 %, so 90 % needs 5 PCs (Kaiser: 3). The variables are only moderately correlated, and Intelligence and FightingSkills are nearly independent of the rest.

• **PC1 = overall physical power** (Captain Marvel, Thor, Hulk vs. Professor X). **PC2 = agile/fast vs. tall/heavy** (Nightcrawler, Shadowcat vs. Juggernaut, Colossus).



Biplot of the standardized superhero data.

PC2 – MNIST digits

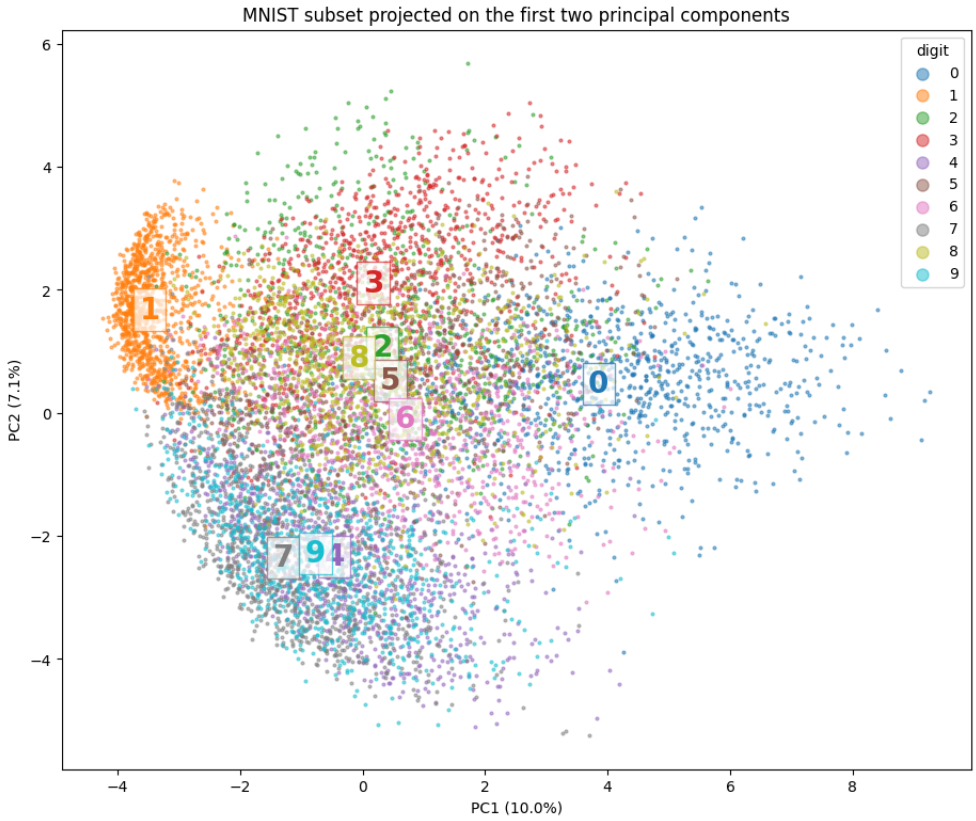
• Rows are images and columns are 784 pixels with gray values 0 – 255. 81 % of the values are 0, and 112 border pixels are always 0.

• **No standardization:** all variables have the same meaning and range. Z-scores would blow up almost-constant border pixels.

• PC1 ≈ 10 %, PC1+PC2 ≈ 17 %. 90 % needs 86 PCs and 95 % needs 151 (of 784), a large reduction but far from 2.

• **2-D plot:** 0 and 1 form their own regions, and PC2 splits 4/7/9 from 2/3/5/8, but within these groups the digits overlap. kNN in 2-D is 42 % accurate, against 95 % with 50 PCs.

• This is expected from (b): 2 PCs keep only 17 % of the variance, and PCA is linear. t-SNE separates the digits much better.



MNIST projected on PC1 and PC2, colored by the true digit.

PC3 – MovieLens movie ratings

• User × movie rating matrix (0.5 – 5 stars), **98.3 % missing**; half of the movies have ≤ 3 ratings.

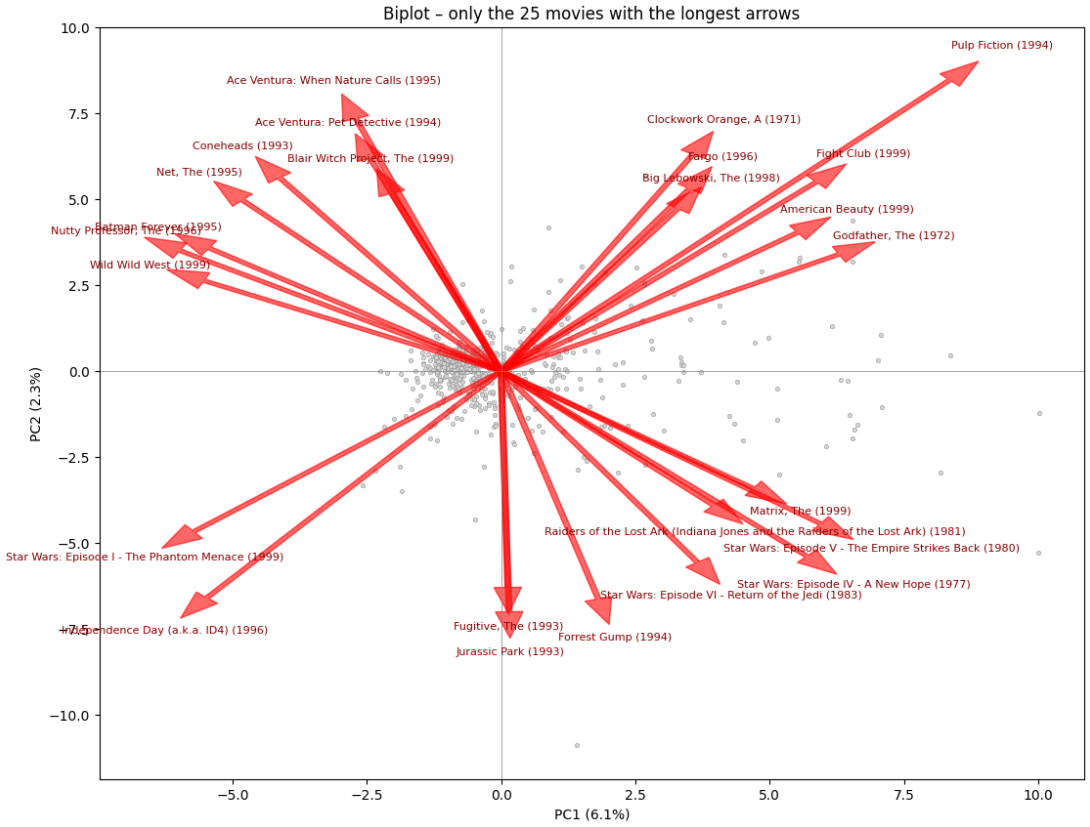
• No column standardization (same scale). Each user is centered by their mean rating and missing values are set to 0, which means 'average for this user'.

• Only 610 users, so at most 609 PCs exist. PC1 ≈ 5 – 6 %, then about 2 % each, with no elbow. 90 % needs ≈ 250 PCs (≈ 180 for the 450 movies with ≥ 50 ratings).

• **PC1 = acclaimed classics vs. weak blockbusters** (Pulp Fiction, Godfather vs. Wild Wild West, Batman Forever). The loadings correlate r = 0.87 with the average rating of each movie.

• **PC2 = edgy/cult vs. mainstream crowd-pleasers** (Clockwork Orange, Fargo, Ace Ventura vs. Jurassic Park, Forrest Gump, Lion King).

• **Limited use:** only 8 % of the variance is shown, the plot is unreadable with all movie names, the users form no groups, and a user's distance from the origin mainly reflects how many movies they rated (r = 0.78).



Biplot of the 450 popular movies, showing only the 25 longest arrows.